



universidade
de aveiro

Sustainable Mobility

Chapter 3 – Sustainable mobility measures



© MCCoelho

Prof.^a Dr. Margarida Coelho

2022/2023

© MCCoelho

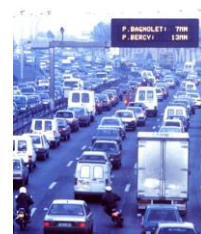
Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho — margarida.coelho@ua.pt



universidade
de aveiro

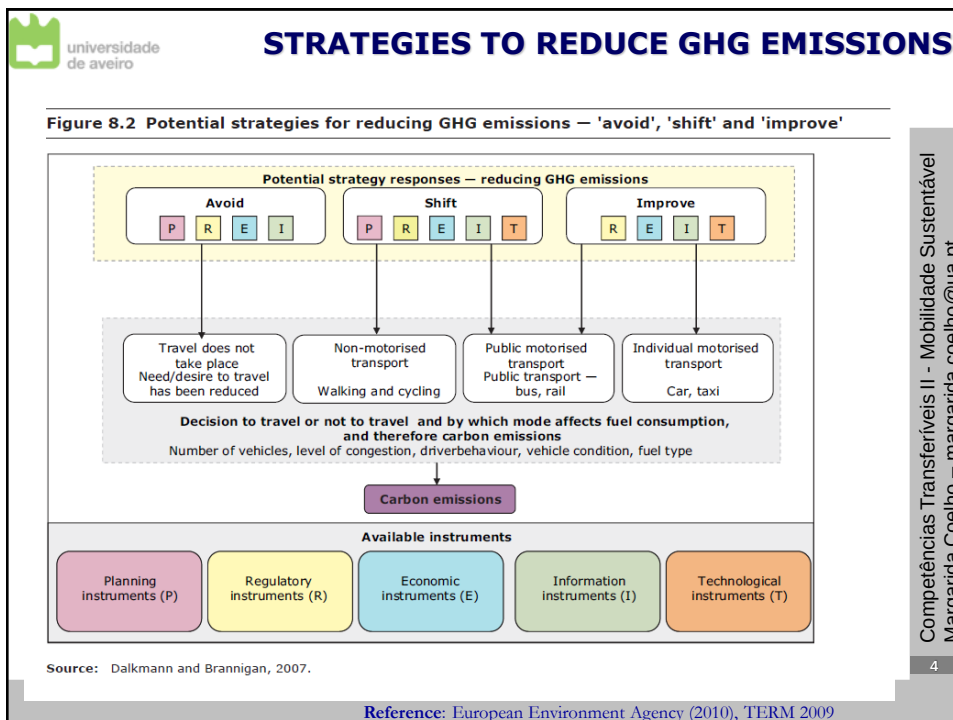
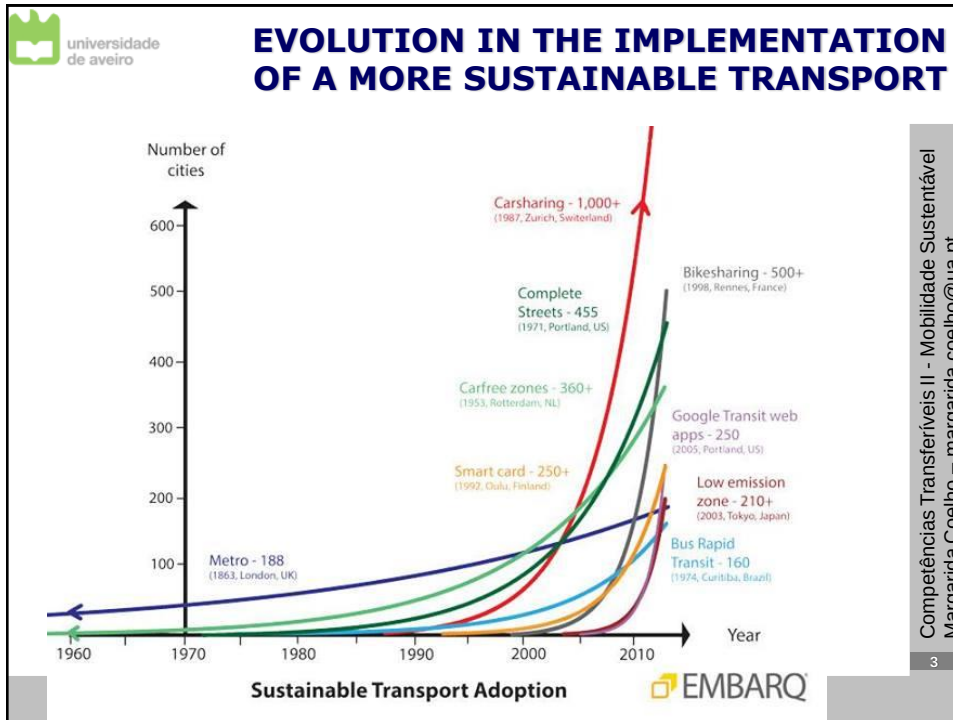
MAIN PROBLEMS OF URBAN MOBILITY

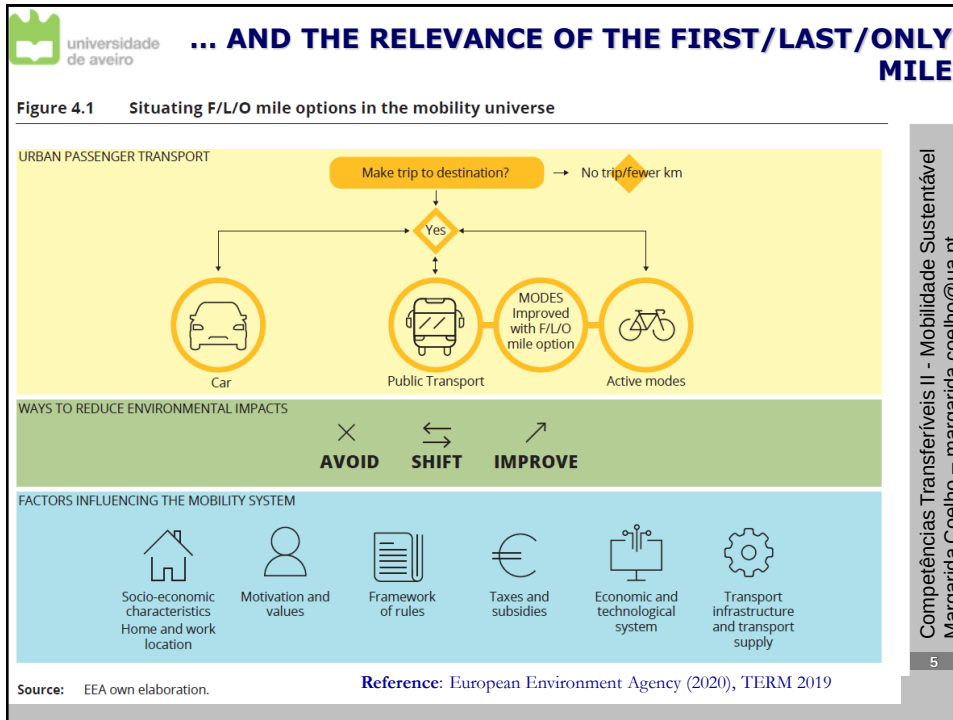
1. Type of travel (not only commuting, e.g. "home → work → home")
2. Tendency for houses construction along highways
3. Need of a multimodal transport :
 - More complex
 - More difficult to assimilate and to promote
 - More barriers to be undertaken
4. Private car is an symbol of freedom...
And travel by car is still very cheap



1 + 2 + 3 + 4 = invitation to use private transport

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho — margarida.coelho@ua.pt





universidade de aveiro

PUBLIC TRANSPORT

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

7

universidade de aveiro

PUBLIC TRANSPORT USE

Main barriers:

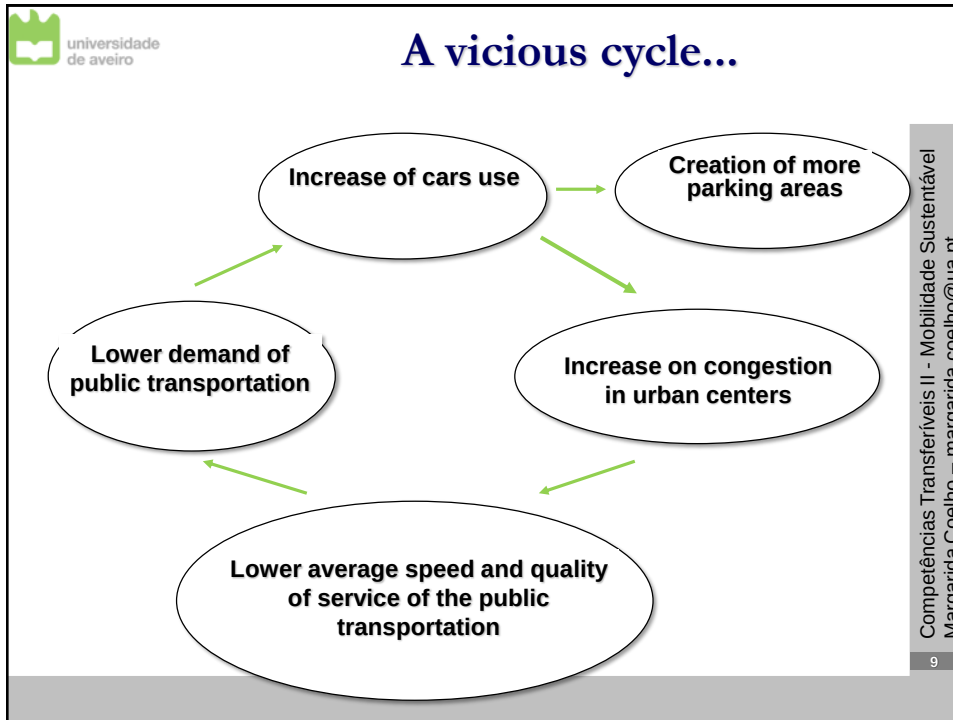
1. Information barriers
2. Physical barriers
3. Psychological barriers
4. Cultural barriers
5. Practical order barriers






Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

8



PUBLIC TRANSPORT USE
⇒ **IMPORTANCE OF A MULTIMODAL INTERFACE**

universidade de aveiro

Transportation services available:

- taxi
- bus
- bicycle
- pedestrian
- train
- boat
- subway

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

10

universidade de aveiro

All images: © MCCoelho

GERMANY...

Abfahrt Departure / Départ

Zeit Time/Tempo	Über via	Ziel Destination	Quitt. Partenza
08:51	10	Regensburg	10
08:51	10	Regensburg	10
08:52	10	Regensburg	10
08:53	10	Regensburg	10
08:53	10	Regensburg	10
08:54	10	Regensburg	10
08:54	10	Regensburg	10
08:54	10	Regensburg	10
08:55	10	Regensburg	10
08:55	10	Regensburg	10

Schickes Auto!

Sparkasse Aachen

11

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho — margarida.coelho@ua.pt

universidade de aveiro

GERMANY...

In the University of Regensburg campus... the importance of information!

Warte auf Karte

Zeit	Linie	Richtung	Haltestelle
20:30	1	Klinikum	Universität
20:30	7	Roter-Brach-Weg	Universität
20:37	6	HBF/Albertstraße	Universität
20:39	11	Wernerwerkstr.	Universität
20:50	21	Burgweinting	Universität
20:51	30	Klenzestraße	Universität
21:00	36	Roter-Brach-Weg	Universität
21:06	36	Klinikum	Universität

OTH-KARTE

Warte auf Karte

Zum Aufbuchen bitte Bildschirm berühren/Please touch the screen to deposit money

Busfahrplan Universität

Zeit	Linie	Richtung	Haltestelle
20:36	6 m	6	Klinikum
20:37	7 m	11	Roter-Brach-Weg
20:39	9 m	C6	HBF/Albertstraße
20:50	20 m	6	Wernerwerkstr.
20:51	21 m	11	Burgweinting
21:00	30 m	4	Klenzestraße
21:06	36 m	11	Roter-Brach-Weg
21:06	36 m	6	Klinikum

The collage features several images related to Geneva and sustainable mobility. At the top left is the logo of the University of Aveiro. Below it is a 'Geneva Transport Card' for 2010-2020, issued by Genève Tourisme and the Bureau des congrès. To the right is a 'ZONE' traffic sign indicating a pedestrian zone with a 20 km/h speed limit. Below the sign is a photo of a tram on a city street. To the right of the tram is a photo of a person pushing a shopping cart on a sidewalk. Below the tram is a photo of a tram with bicycles parked on it. To the right of the tram is a photo of a tram on a city street with a cyclist in the foreground.

STOCKHOLM...

... the importance of information and of "small-big" details detalhes...



Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

15

MINIMETRO IN PERUGIA...



Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

 **FREE PUBLIC TRANSPORTATION AT THE CITY CENTRE**

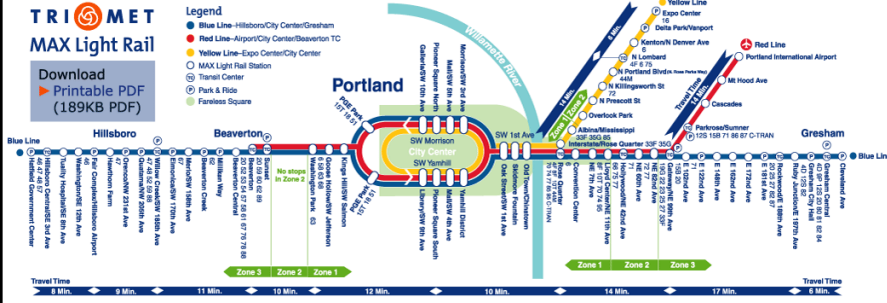
Ex. Portland (Oregon, USA)

TRI MET MAX Light Rail

Download
Printable PDF
(189KB PDF)

Legend

- Blue Line-Hillsboro/City Center/Gresham
- Red Line-Airport/City Center/Beaverton TC
- Yellow Line-Expo Center/City Center
- MAX Light Rail Station
- Transfer Center
- Park & Ride
- Transfer Square



Portland

Travel Time

8 Min. 9 Min. 11 Min. 13 Min. 12 Min. 10 Min. 14 Min. 17 Min. 8 Min.

Zone 1 Zone 2 Zone 3





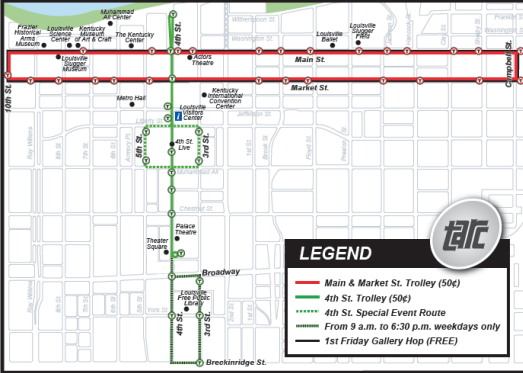
© MCCoelho

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho — margarida.coelho@ua.pt

17

 **...OR VERY CHEAP!**

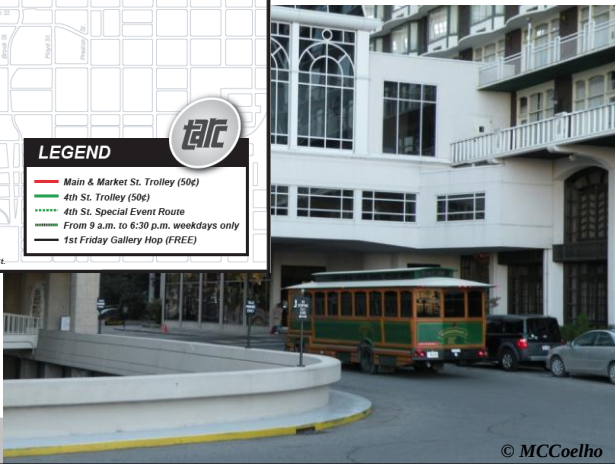
Ex. Trolley in Louisville, Kentucky, USA... 0.5 USD per trip



LEGEND

- Main & Market St. Trolley (50¢)
- 4th St. Trolley (50¢)
- 4th St. Special Event Route
- From 9 a.m. to 6:30 p.m. weekdays only
- 1st Friday Gallery Hop (FREE)

Other examples in USA:



© MCCoelho

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho — margarida.coelho@ua.pt

18

universidade de aveiro

The contribution of Walking for sustainable mobility



© MCCoelho

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

19

universidade de aveiro

~2000 years ago...

Pompeii



Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

20





universidade
de aveiro

ACCESSIBILITY PLAN OF LISBON

PLANO
DE ACESSIBILIDADE
PEDONAL

LISBOA
Câmara Municipal





- Implemented in Alvalade and Campo de Ourique
- “Traffic snake game”

A Pé para a Escola... em Segurança!

Aplicação dos Princípios de Acessibilidade e Desenho Inclusivo às Estratégias de Segurança Rodoviária nas Escolas do 1.º Ciclo do Ensino Básico




Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho — margarida.coelho@ua.pt

23





universidade
de aveiro

CASE STUDIES - PARIS

Urban planning serving the quality of life of citizens



© MCCoelho



© MCCoelho

CASE STUDIES - NYC

Urban planning serving the quality of life of citizens

© MCCoelho



Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho — margarida.coelho@ua.pt

25

CASE STUDIES - NYC

Urban planning serving the quality of life of citizens



© MCCoelho

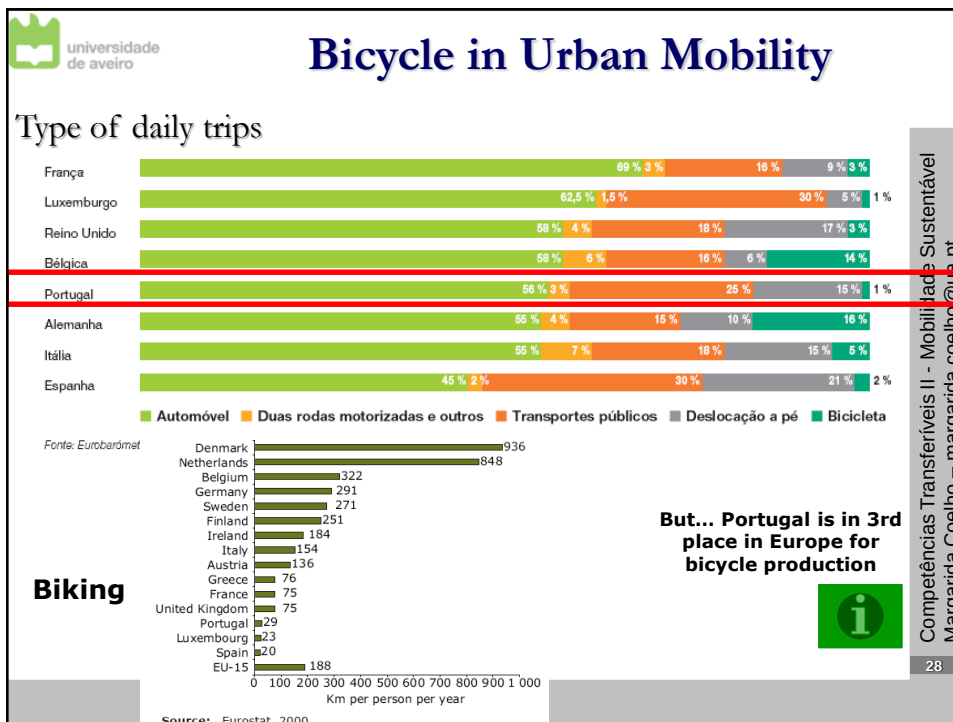
universidade de aveiro

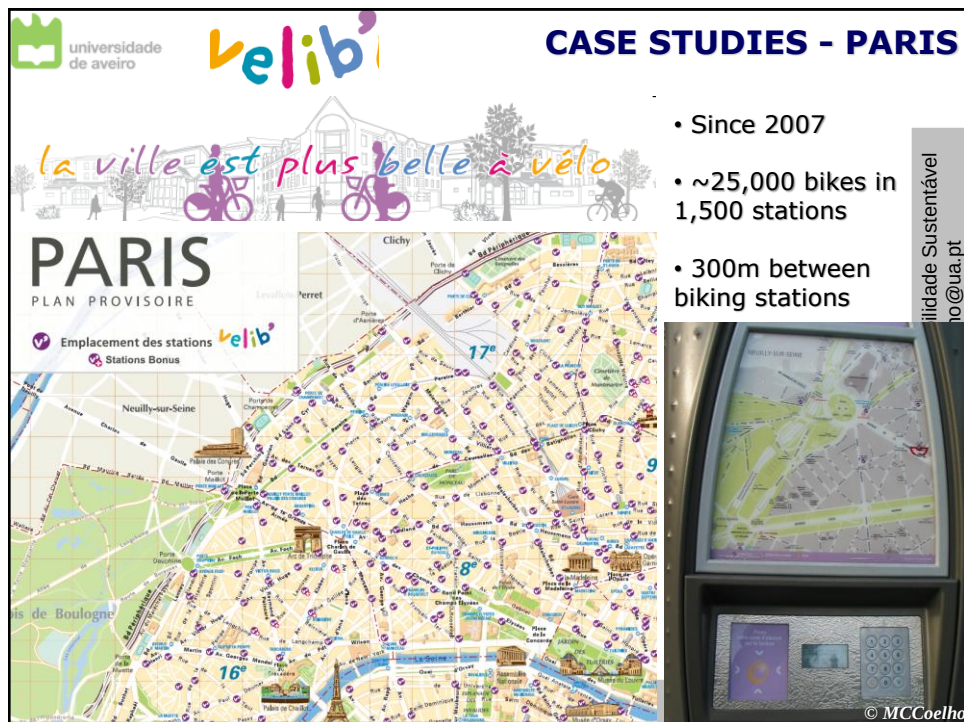
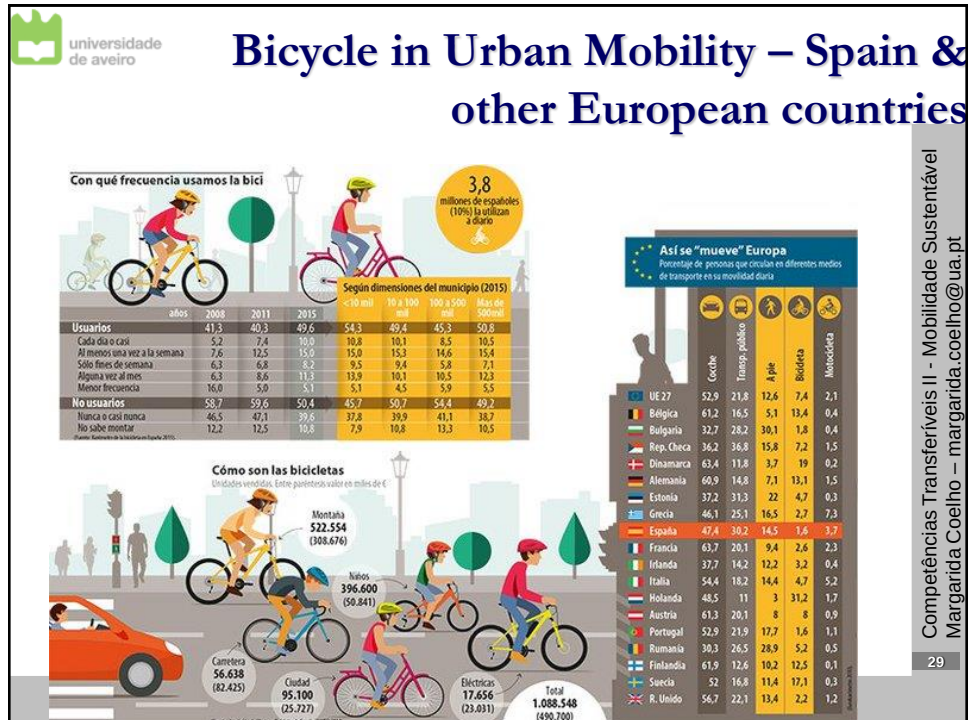
The contribution of Biking for sustainable mobility

© MCCoelho

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

27





CASE STUDIES - PARIS



- Service 24/7
- 1 day, 7 days, 1 year... 1st half hour free
- Problems: accidents, lack of compliance with the road regulations, vandalism, robbery



CASE STUDIES - LYON

Not only bike sharing, also
trotinette-sharing!



CASE STUDIES - LYON

The most important: use the right mix!



CASE STUDIES - BELGIUM

All images: © MCCoelho



Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

34

 universidade de aveiro

CASE STUDIES - BELGIUM

All images: © MCCoelho






Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho — margarida.coelho@ua.pt

35

 universidade de aveiro

CASE STUDIES – THE NETHERLANDS

- Bicycle as a relevant transport mode in the intermodal station of Delft
- It is trendy to bike!

© MCCoelho

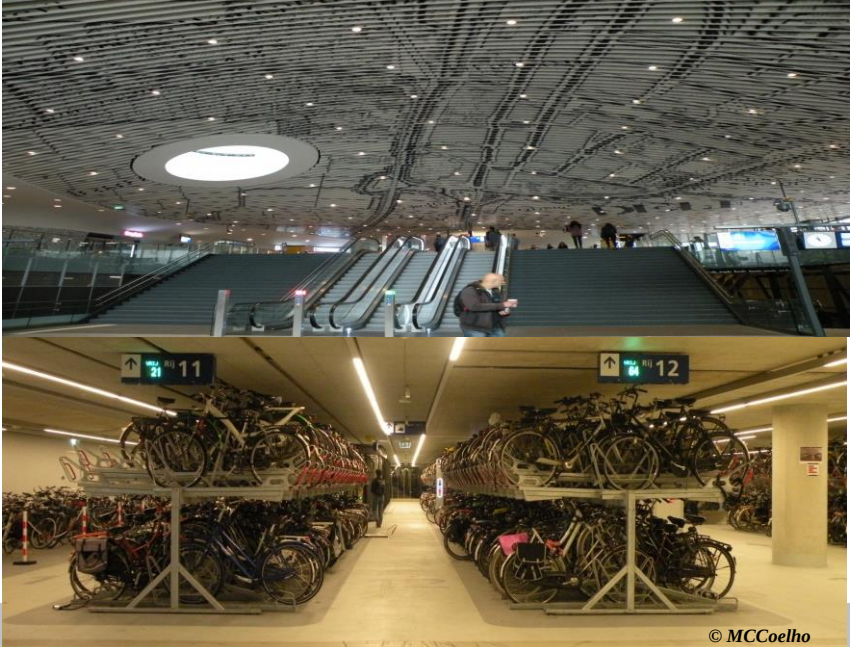


Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho — margarida.coelho@ua.pt

36

universidade de aveiro

CASE STUDIES - HOLLAND



Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

37

© MCCoelho

universidade de aveiro

CASE STUDIES - HOLLAND



De maximale stallingsduur in deze stalling is 14 dagen

14
DAGEN
GRATIS

Fietsenstalling met toezicht

- Stallen
- Fiets & Service
- OV-fiets

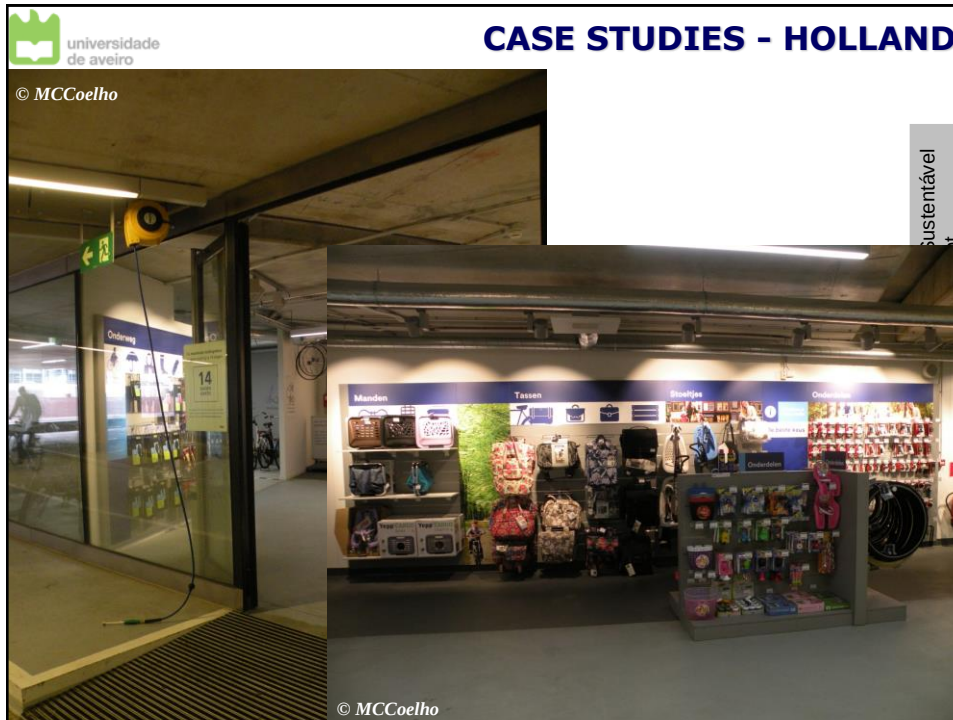
Openingstijden Stalling

24 uur per dag open

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

38

© MCCoelho



 universidade de aveiro

CASE STUDIES - COPENHAGEN

- > arterial of Copenhagen (Nørrebrogade, 14.3 km): closed to private cars (exclusive to bikes and public transportation) → 33000 cyclists/day
- 36% of the 500 000 inhabitants of Copenhagen go to work by bike since “it is a cheap, easy transport mode and a good way to transport both goods and even children”.
- The Municipality wants to enlarge the traffic closure to the other 2 biggest arterials of the city: Amagerbrogade e Østerbrogade ⇒ Objective: half of the population travels by bike in 2015.
- ~400 km bike lanes



Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

40

© MCCoelho



 universidade de aveiro

CASE STUDIES - EUROVELO (European cycle route network)

- Objective: to develop 12 long distance cycle routes along Europe
- Total length: 60,000 km
- Characteristics of “EuroVelo” routes:
 - Slope < 6%
 - Width enough for 2 cyclists
 - Average TMD < 1,000 vehicles
 - Have protection in + 80% of the total length
 - To be open every day
 - Supplying stations every 30 km
 - Lodging every 50 km
 - Public transport availability every 150 km





 universidade de aveiro

CASE STUDIES – RIO DE JANEIRO, BRAZIL

All images: © MCCoelho





Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

44



CASE STUDIES – AVEIRO REGION

universidade de aveiro

lifecycle
bicicleta e vida

Seminário Saúde e Bicicleta
Life Cycle: a Bicicleta e Vida
7 Abril 2009 (9.30 h-18.00 h)
Centro Congressos Aveiro

Para a escola com pedalada.

BUGA

CICLORIA

- Municipalities of Estarreja, Ovar, Murtosa + UA.
- Objective: To develop a culture of use of soft mobility modes in the Ria de Aveiro region
- System of cycle paths (120 km) + bike system offering collective use (100 / municipality) + bike lanes themed tours

Competências Transversais
Margarida Coelho

46



CASE STUDIES – AVEIRO REGION

- Relevant projects involving bike and cycling mobility (Cycle Route Ria - Polis , BE Agueda, Sangalhos Velodrome , bike lanes and various infrastructures)
- Region where in Portugal more people regularly cycle - 8 × > national average - 3.9% vs. 0.5 % (INE , 2011)
- It is in this region that is the county with > n.º users in relative terms (Murtosa to 17%), and 5 municipalities with > value in absolute terms , Ílhavo , Aveiro, Estarreja , Ovar and Murtosa , with 2,160 , 1,351 , 996, 935 and 893 users of the bicycle, respectively.
- More than half the population has a bike at home (535 bicycles / 1000 inhabitants), value > average motorization rate of municipalities - 502 veh / 1,000 inhabitants.

Competências Transferíveis II - Mobilidade Sustentável
 Margarida Coelho – margarida.coelho@ua.pt

47



CASE STUDIES – AVEIRO REGION

- On average, 29% of people admit to use the bike in their daily travel (values range from 7% in Sever do Vouga to 37% and 42% in Murtosa and Ílhavo, respectively), being dependent on the improvement and expansion of the cycling network (57%), improving the conditions of the pavements (17%),> road safety (6%) and <road traffic (2%) (PIMTRA, 2012)
- The region of Aveiro has unique conditions for the promotion of cycling tourism as an activity that mobilizes visitors / tourists moving by bike, on vacation or leisure activity that has had a growing importance at European level
-

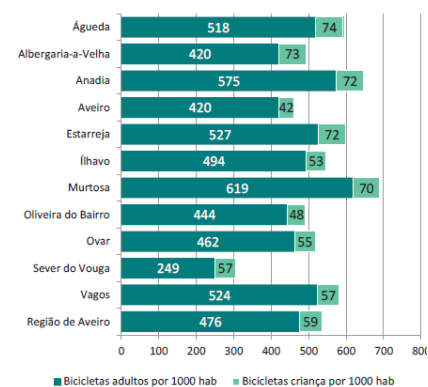
Competências Transferíveis II - Mobilidade Sustentável
 Margarida Coelho – margarida.coelho@ua.pt

48

- Promotion of Electric and Regular Bikes in Academic Communities
- 15 academic institutions involved
- 237 bicycles to the University of Aveiro

CASE STUDIES – AVEIRO REGION

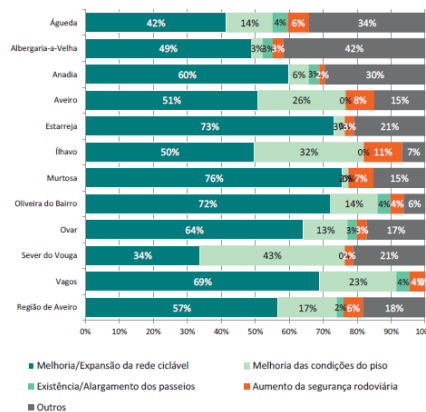
Figura 131 – Bicicletas por mil habitantes por concelho



Fonte: Inquérito à Mobilidade 2011/2012

Ref: Plano Intermunicipal de Mobilidade e Transportes da
Região de Aveiro, 2012.

Figura 133 – Condições necessárias para uma maior utilização da bicicleta nas deslocações quotidianas



Fonte: Inquérito à Mobilidade 2011/2012



universidade
de aveiro

CASE STUDIES - LISBON

- 80 km of bike lanes in 2009
- There is no quantification on the number of cyclists in Lisbon
- Plan: electric bike sharing
- Map Lisboa Ciclável: <http://lisboaciclavel.cm-lisboa.pt/>






Competências Transferíveis II - Mobilidade Sustentável

Margarida Coelho — margarida.coelho@ua.pt

51



universidade
de aveiro

CASE STUDIES - LISBON






Competências Transferíveis II - Mobilidade Sustentável

Margarida Coelho — margarida.coelho@ua.pt

52

universidade de aveiro

Road conflicts between cyclists and pedestrians

© MCCoelho



Mobilidade Sustentável
coelho@ua.pt



© MCCoelho

universidade de aveiro

Biking - Conclusions

- Return the street to citizens (and not to road vehicles)
- Transport as a social inclusion factor
 - Specific groups such as children, youth, women, the elderly and people with reduced mobility depend more often of a combination of walking, cycling and public transport
 - We must study and listen what they think about the urban environment and the transport system

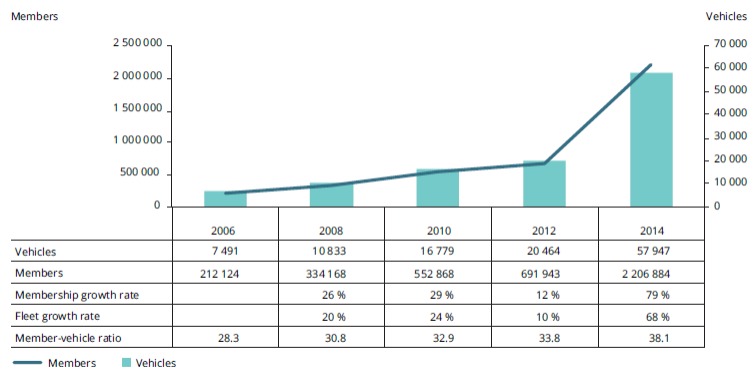



© MCCoelho

OTHER MEASURES

"CAR SHARING"

Figure 7.1 European trends in car sharing



Source: Shaheen and Cohen, 2016.

Exemplos:

<https://www.drive-now.com/pt/pt/lisbon/>
<https://www.citydrive.pt/index.php/en/home/>

universidade de aveiro

"CAR POOLING"

Portugal está cheio de lugares vazios.
ADIRA JÁ AO GALPSHARE®

galpshare®
partilhe a sua energia positiva

Take the stress out of the journey, car share

CAR POOLING

ORIGEM 1

ORIGEM 2

ORIGEM 3

DESTINAÇÃO 2

DESTINAÇÃO 1

57

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho — margarida.coelho@ua.pt

universidade de aveiro

"PARK & RIDE"

MASS PINE PARK & RIDE

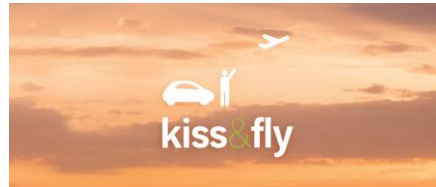
P+ Park and Ride

58

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho — margarida.coelho@ua.pt

A VARIATION OF "PARK & RIDE": "KISS & RIDE" ou "KISS AND FLY"

Zone for exit/entry of passengers

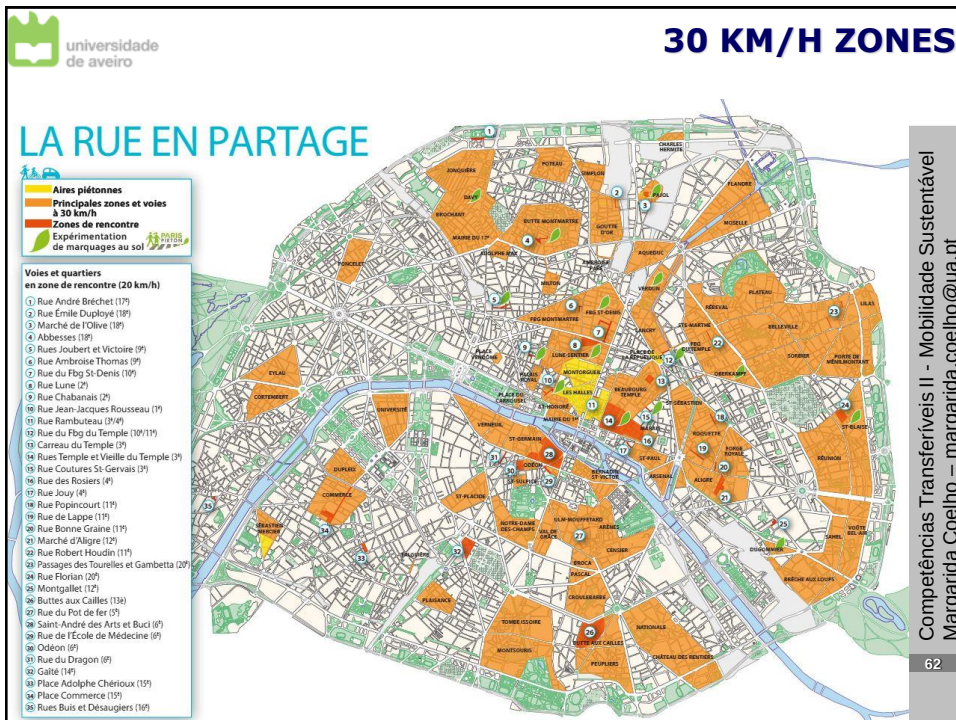


Lisbon Airport - 1st 10 minutes: free

TRAFFIC CALMING



© MCCoelho



universidade de aveiro

HOV LANES





Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho — margarida.coelho@ua.pt

63

universidade de aveiro

LOGISTICS – GENEVA & LAUSANNE...




© MCCoelho

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho — margarida.coelho@ua.pt

64

universidade de aveiro

VEHICLES IN HISTORICAL CENTRES – EX. CÓRDOBA & GRANADA...



© MCCoelho



© MCCoelho

Mobilidade Sustentável
coelho@ua.pt

universidade de aveiro

VEHICLES IN HISTORICAL CENTRES – EX. LYON



© MCCoelho

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

66

 universidade de aveiro

PARKING POLICIES

To decrease the number of vehicles in the city centre:

Núcleo Central (10 Km)	• ↑ Estacionamento tarifado (via pública + parques) • ↑ Efeito dissuasor uso TI • ↓ Oferta estacionamento na via pública e parques • ↑ “P+R” nos limites do Núcleo Central e do hipercentro • Recuperação e requalificação dos eixos viários urbanos
Coroa Transição (10-30 Km)	• ↑ Estacionamento “P+R” (gratuito ou pago c/ direito ao comboio e a TC do tipo expresso) • ↑ “P+R” ao longo dos eixos mais congestionados • Completar sistema rodoviário radio-concêntrico • Desenvolver malha rodoviária secundária, incluindo variantes aos centros urbanos • Recuperação e requalificação dos eixos viários urbanos
Pólos Secundários (+ 30 Km)	• ↑ Estacionamento “P+R” (gratuito ou pago c/ direito TC) • Política de estacionamento nos centros • Recuperação e requalificação dos eixos viários urbanos • Completar variantes aos centros urbanos

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho — margarida.coelho@ua.pt

67

Fonte: Prof. F. Nunes da Silva

 universidade de aveiro

PARKING - EXAMPLES...



© MCCoelho



Ref: http://www.roadtraffic-technology.com/contractors/driver_info/dambach/dambach1.html

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho — margarida.coelho@ua.pt

68



universidade
de aveiro

A SOLUTION: "CONGESTION CHARGING"



All images: © MCCoelho

270 video cameras
£11.50 / entry




Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho — margarida.coelho@ua.pt

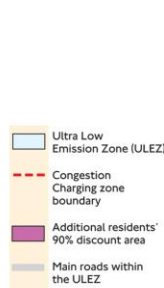
69



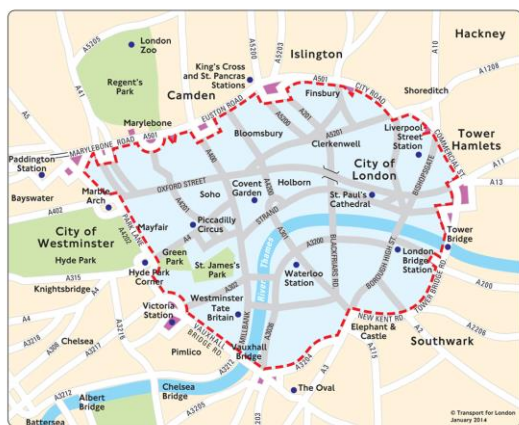
universidade
de aveiro

A SOLUTION: "CONGESTION CHARGING"

Ultra Low
Emission Zone



- Ultra Low Emission Zone (ULEZ)
- Congestion Charging zone boundary
- Additional residents' 90% discount area
- Main roads within the ULEZ



2003 in London City: "Congestion Charge" using a toll monitoring system:

- Traffic flow improved by 37%
- Traffic volume reduced by around 40% in city toll charge periods

More pollutant vehicles pay a higher tax: £11.50 + 10!

«London's £10 T-charge comes into effect in fight against toxic car fumes

Drivers of older, more polluting petrol and diesel cars in centre of capital now liable for fee on top of congestion charge»

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

70

universidade de aveiro

Z.T.L. AT PERUGIA...

All images: © MCCoelho

universidade de aveiro

"LOW EMISSION ZONES" (Z.E.R.) IN LISBOA



ZER:

área em que a circulação de veículos **mais poluentes** / mais antigos **é restringida**

≠



Taxa de Congestionamento :
[Congestion Charge]

área na qual a **circulação** (seja de que veículo for) **é paga** (mas **permitida**)

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

72

Ref: F. N. Silva, C.M. Lisboa

**universidade
de aveiro**

"LOW EMISSION ZONES" (Z.E.R.) IN LISBOA

Compências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

Momentos Rápidos - Mais Oxigénio - Respirar Melhor
Viver Mais Lisboa

O que é? Uma Zona de Emissões Reduzidas (**ZER**) é uma zona onde a circulação de veículos é condicionada aos veículos menos poluentes por razões de proteção do meio humano e o cumprimento das legislações comunitária e nacional.

História: Diretiva única, desde 86 à 2001.

Como vai funcionar? São as veículos que respeitem no máximo os normas de emissões Euro 1 introduzidos após ano de construção é posterior a julho de 1992) é que poderão circular nesta Zona.

Onde?

- Av. da Liberdade/Baixa
- Eixo Verde
- Residentes em Lisboa;
- veículos de emergência;
- especiais e de passagens com mobilidade reduzida;
- veículos de interesse histórico;
- abandonados certificados e transporte público.

ZER
Zona de Emissões Reduzidas

Av. da Liberdade/Baixa [1ª Fase]
circulação interdita a veículos anteriores a Julho 1992

Exceto Residentes,
Transportes Públicos,
Veículos Históricos

ZER
Zona de Emissões Reduzidas

mais Lisboa

Menos Poluição
Mais Oxigénio
Respirar Melhor
Viver Mais Lisboa

foi aprovado em Reunião de Câmara, no dia 18 de Maio a criação de uma Zona de Emissões Reduzidas na Cidade de Lisboa - ZER (proposta nº 247/2011 publicada no 3º suplemento ao Boletim Municipal nº900, de 19 Maio 2011).

A primeira fase caracteriza-se pela restrição à circulação de veículos que não respeitem as normas de emissão EURO 1 (veículos fabricados antes de Julho 1992), no eixo da Av. da Liberdade (a sul da Av. Alexandre Herculano) e a Baixa.

O Município de Lisboa está desenvolver o estudo respeitante à 2ª Fase da ZER, cuja implementação entrará em vigor em Janeiro de 2012, alargando os seus limites de intervenção e exigência. Um passo importante para se dar início a um processo de melhoria da qualidade do ar.

zona
até maio 86 ou 2001
até Euro 1
até anterior a julho/92
exceto residentes em Lisboa
transporte público
veículos históricos

Ref: C.M. Lisboa

Av. da Liberdade/Baixa [1ª Fase]
circulação interdita a veículos anteriores a Julho 1992
(Exceção Residentes em Lisboa, Transportes Públicos, Veículos Históricos)

73



universidade
de aveiro

"LOW EMISSION ZONES" (Z.E.R.) IN LISBOA





Proposta 1ª Fase da ZER: eixo Av. Liberdade/Baixa



a) 2ª a 6ª feira, das 8:00 às 20:00

b) Aplicável a **ligeiros** e **pesados** (gasolina e diesel)

c) Apenas podem circular viaturas que respeitem a norma de emissões poluentes **Euro 1** ou **superior**

d) **Exceções admitidas:**

- Transporte Público
- Veículos de Emergência e especiais
- Veículos Históricos
- Residentes no local

Ref: F. N. Silva, C.M. Lisboa

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

74

 universidade de aveiro

"LOW EMISSION ZONES" (Z.E.R.) IN LISBOA – 2ND PHASE

Período de funcionamento da proibição na Zona 2:
Dias úteis: das 7h00 às 21h00

Como identificar as zonas abrangida pela proibição:
As entradas nas ZER encontram-se identificadas através de sinalização vertical existente no local.

zona



dias úteis 8h às 20h

pré-Euro 1
veic. anter. a julho/92

excepto
resid., transp. públicos
veículos históricos



Limite 2ª Fase ZER:
(ZONA 2)
Av. Ceuta | Eixo Norte-Sul
Av. Forças Armadas | Av. EUSA
Av. Marechal António Spínola |
Av. Santo Condestável |
Av. Infante D. Henrique

Limite 1ª Fase ZER
(ZONA 1)

75

Ref: ANTRAM, 2012.

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

 universidade de aveiro

"LOW EMISSION ZONES" (Z.E.R.) IN LISBOA – 3RD PHASE

- From January 2015, prior to 2000 vehicles will no longer be able to move on Avenue Liberdade, in downtown Lisbon (7h-21h, weekdays).
- Also the January 15, 2015, vehicles built before 1996 will no longer be able to move within the area bounded by Avenue Ceuta, Avenue of the Armed Forces and the United States Avenue (Zone 2 ZER) (7h-21h, weekdays).
- Exceptions list: Emergency vehicles and persons with disabilities, historical, belonging to residents in Lisbon, natural gas, LPG and motorcycles, police, military, prisoner transport, armored transport values and taxis.
- Taxis will continue to be treated differently: from July 2015 transport vehicles from previous passengers to 1992 will not flow both in Zone 1 and 2. 1 year later, those which have been built by 1996, will no longer travel on the Avenue Liberdade and downtown. Only in July 2017 is that taxis will be subject to the same restrictions will be imposed on cars and trucks next year, because of the difficulties that some professionals have been expressing for the purchase of latest vehicle registration.

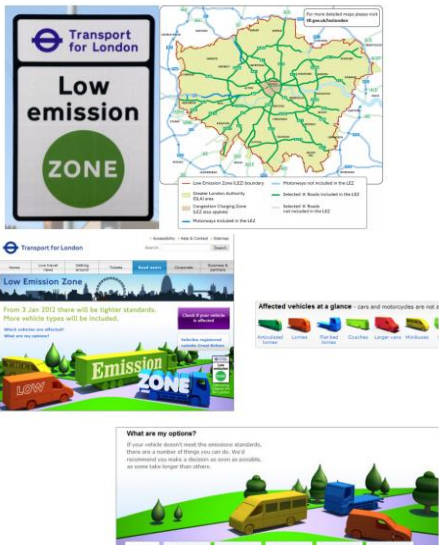
76

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

 universidade de aveiro

LOW EMISSION ZONES

Londres (apenas pesados):
desde 2008



Berlim (ligeiros e pesados):
desde 2008



Other examples:
<http://www.lowemissionzones.eu/>

Ref: F. N. Silva, C.M. Lisboa

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho — margarida.coelho@ua.pt

77

 universidade de aveiro

EUROPEAN MOBILITY WEEK / DAY WITHOUT CARS



EXPLORE ALL YOUR TRAVEL OPTIONS

YOU CONTROL CLIMATE CHANGE.

WALK, CYCLE, GO BY BUS, CAR SHARE, CARPOOL.

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho — margarida.coelho@ua.pt

78

 **CIVITAS**

- CIVITAS - cleaner and better transport in cities – “City-VITAlity-Sustainability”.
- CIVITAS I: beginning of 2002; 19 cities
CIVITAS II: beginning of 2005; 17 cities
CIVITAS PLUS: beginning of 2008 – 2012; 25 cities
- Main objective: Promote and implement sustainable mobility measures
- Portuguese cities:
Porto, Coimbra, Funchal, Torres Vedras




Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

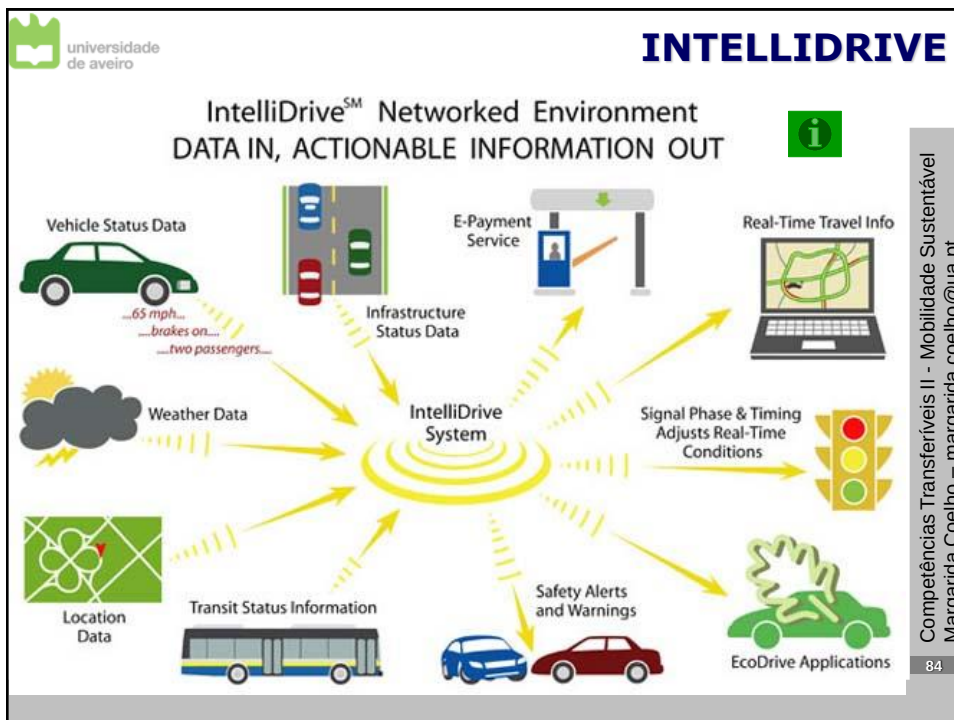
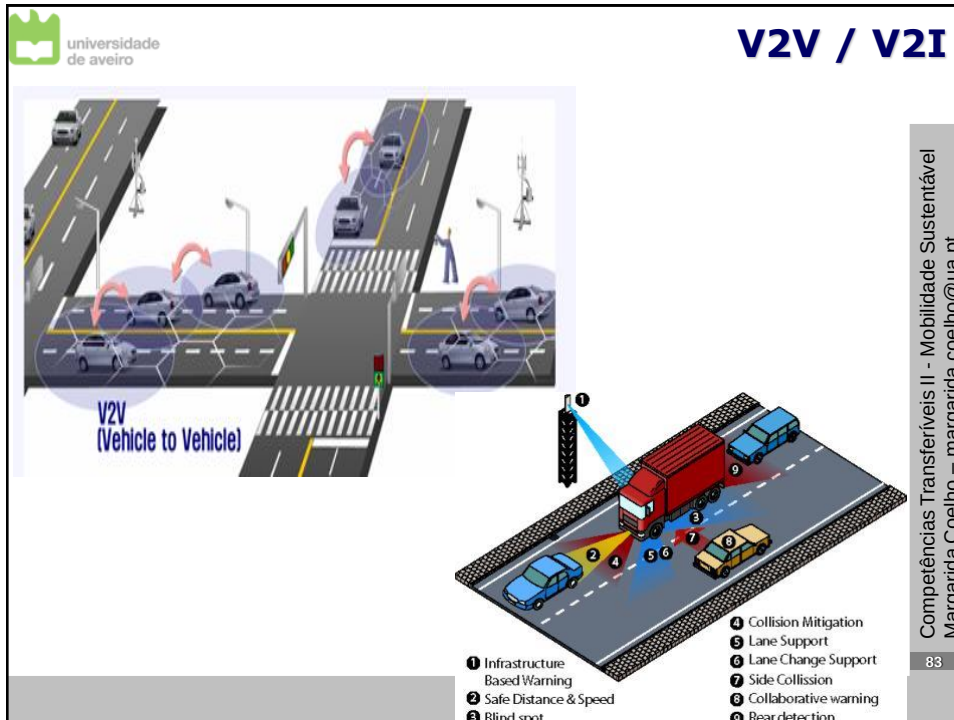
79

 **DIGITALIZATION IN MOBILITY**



Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

80



Inspiration in Nature... and people...



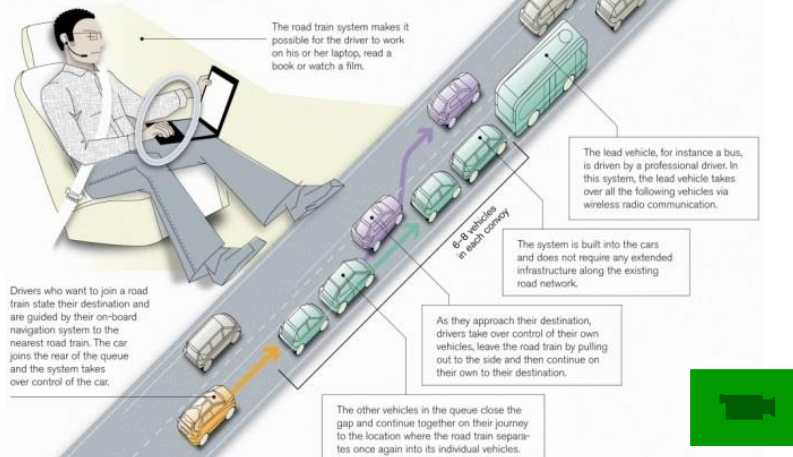
What if we connect vehicles?!?

V2V/V2I $\Rightarrow$ Connected vehicles $\Rightarrow$ Autonomous vehicles

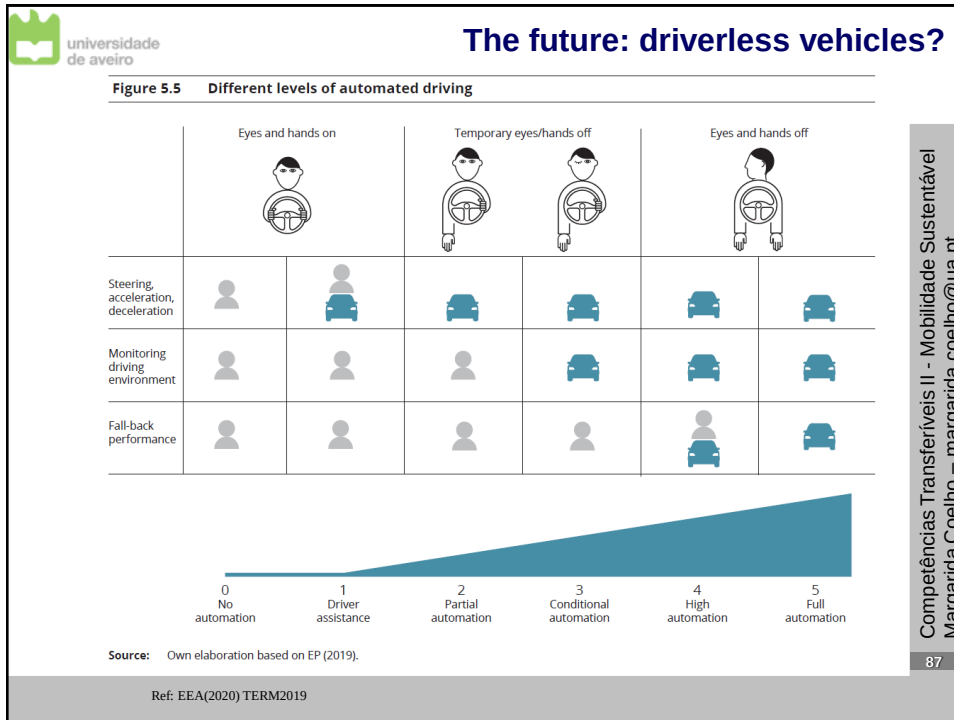
Ex. Project “Road Trains for the Environment” (Sartre) – “Road Trains”

Join a road train

A safe and energy-efficient way to travel



Vehicles circulate interconnected by sensors, establishing a kind of train.



There is already some autonomy on the road! → Driver assistance systems

- Alert the driver
- Active assistance to prevent the accident or mitigate its consequences

They also help drivers:

- To maintain a safe / steady speed
- To maintain safety distance with other vehicles

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

88

 universidade de aveiro

The future: Electric driverless vehicles?

Ref: Prestige Electric Car

Ref: Inhabitat

Ref: Government Technology

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho — margarida.coelho@ua.pt

89

 universidade de aveiro

MaaS – Mobility as a Service

Figure 5.4 Levels of MaaS integration

Higher cognitive user effort			Lower cognitive user effort		
Level 0	Level 1	Level 2	Level 3	Level 4	Level 5
No integration: no operational, informational or transactional integration across modes	Basic integration: informational integration across (some) modes	Limited integration: informational integration across (some) modes with some operational integration and/or transactional integration	Partial integration: some journeys offer a fully integrated experience	Full integration under certain conditions: some but not all available modal combinations offer a fully integrated experience	Full integration under all conditions: full operational, informational and transactional integration across modes for all journeys

Notes: Cognitive user effort: the effort involved in relying upon the mobility system beyond the private car to fulfil mobility goals; operational integration: interchange penalties are low and door-to-door journey experience is 'seamless'; informational integration: journey planning and execution information for available modes is offered through one interface; transactional integration: payment and any required booking and ticketing is offered through one interface.

Source: Lyons et al. (2019).

Ref: EEA



ECODRIVING

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

91



What is Ecodriving?

A form of efficient driving that reduces fuel consumption, pollutant emissions and greenhouse gases, and road accidents (the hard accelerations and brakings decrease) .



Links :

www.ecoconducao-portugal.pt

http://ec.europa.eu/dgs/energy_transport/

<http://www.ecodrive.org>

<http://www.treatise.eu.com> <http://www.greener-driving.net>

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

92



universidade
de aveiro

Ecodriving – Pratical suggestions

- The **choice of car** will have significant implications in terms of consumption , emissions and safety. On car purchase, also take into account the type of fuel, power, information on fuel consumption and emissions, aerodynamics and all indications on the efficiency and vehicle maintenance. In new vehicles, fuel consumption can vary between 4 and 15 liters/100 km. As for CO2 emissions from new vehicles, these can vary between 115 and 280 g/km .

Competências Transferíveis II - Mobilidade Sustentável
 Margarida Coelho – margarida.coelho@ua.pt

93



universidade
de aveiro

Ecodriving – Pratical suggestions

- **Turn on the car engine** only immediately before the trip and turn off the car where it is stopped more than a minute.
- Lead to the most **constant speed** possible and drive smoothly, avoiding accelerations / decelerations (Ex. saves up 15 % of fuel driving at 80 km/h instead of 100 km/h) .

Competências Transferíveis II - Mobilidade Sustentável
 Margarida Coelho – margarida.coelho@ua.pt

94



Ecodriving – Pratical suggestions

- Using the highest possible **speed gear** $\Rightarrow$ high speed gear results in lower rotation $\Rightarrow$ $<$ Fuel consumption (saving potential: 10%)
- Comply with **speed limits**, achieving fuel economy and contributing to road safety (an increase of 10% in speed can cause a 15% increase in fuel consumption)

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

95



Ecodriving – Pratical suggestions

- In steep descents, the vehicle should keep a **compatible gear** (with the engine), obtaining better safety and lower consumption .
- **Adjust the vehicle speed** to avoid traffic and changing traffic lane.
- Choose the **best route** for journeys and try to anticipate traffic flow . A well planned trip is a nice feature to save fuel (savings potential: 5%).

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

96



universidade
de aveiro

Ecodriving – Pratical suggestions

- **Avoid carrying luggage** in the car's roof (to change the aerodynamic characteristics of the vehicle increases by about 5 % fuel consumption)
- **Remove all unnecessary weight** from the trunk or back seats $\Rightarrow$ Savings potential : 3%
- **Avoid** using the car for short journeys.

Competências Transferíveis II - Mobilidade Sustentável
 Margarida Coelho – margarida.coelho@ua.pt

97



universidade
de aveiro

Ecodriving – Pratical suggestions

- **Use the air conditioner only when necessary.**
The air conditioning system may consume up to half a liter of fuel per hour and at the beginning of each trip may represent an increase in consumption of about 10 %.
- **Keep vehicle tuned** and check the oil level regularly $\Rightarrow$ Potential savings : 3%

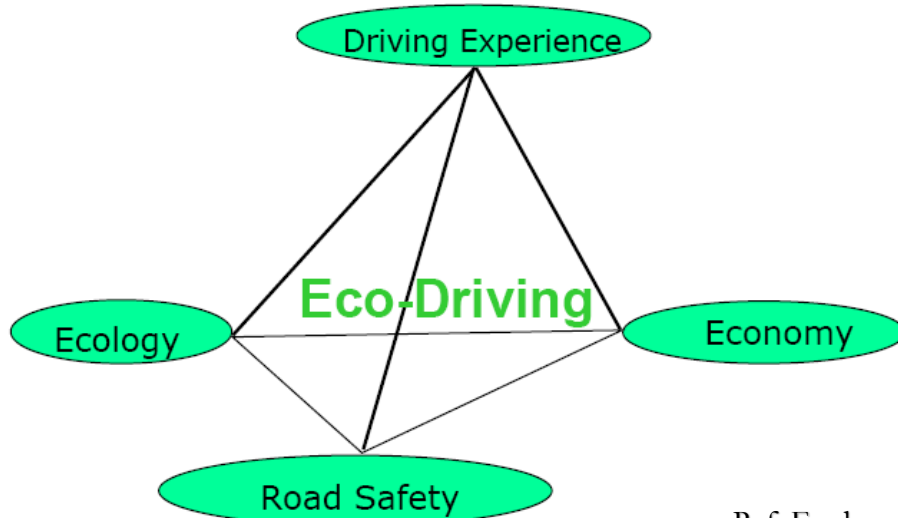
Competências Transferíveis II - Mobilidade Sustentável
 Margarida Coelho – margarida.coelho@ua.pt

98

Ecodriving – Practical suggestions

- **Check every month the tire pressure.** A wrong pressure forces the anticipated replacement of tires. A too low pressure increases the rolling resistance (wear side) and fuel consumption. Too high pressure causes wear in the tire center and a smaller grip on driving (the correct pressure allows savings of up to 3%).

To manage this trade-off...



Ref: Ford

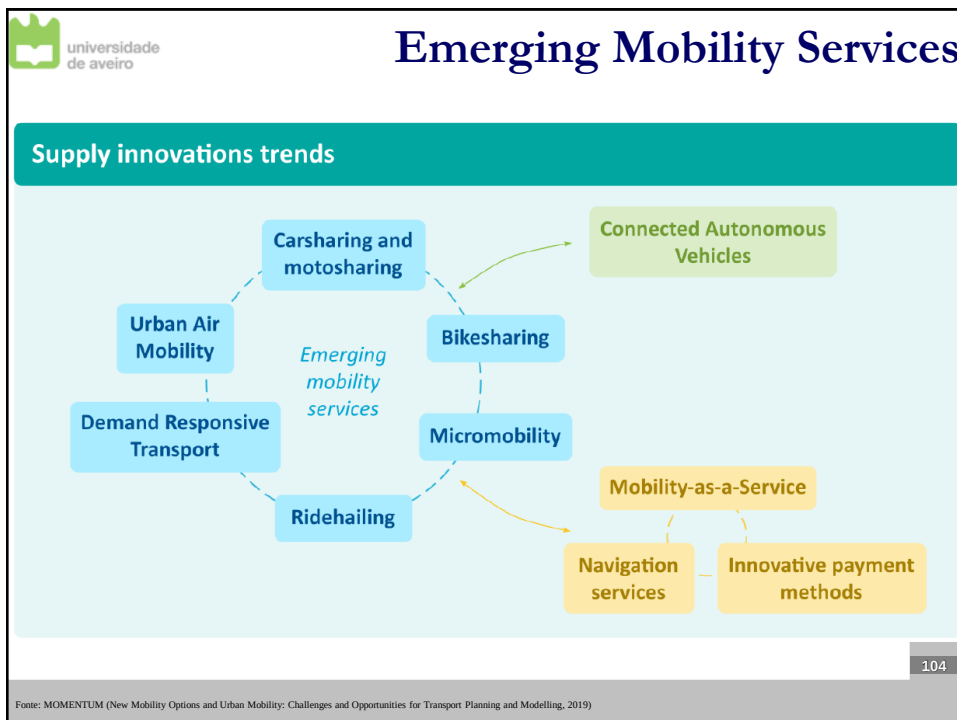
Ecodriving – Conclusions

- Synergy between Eco- Driving & Technology
- Various opportunities and challenges
- According to a European survey, citizens have small knowledge about existing technology $\Rightarrow$ importance of dissemination and proper communication
- Technology and behavior of ecodriving by drivers (mainly of conventional vehicles) are complementary $\Rightarrow$ Strategic importance for reducing CO2 emissions in the transport sector
- Technology is important, but the main (short-term effect) must be done by drivers $\Rightarrow$ major importance of driving schools

CONCLUSIONS & DISCUSSION



© MCCoelho



 universidade de aveiro

OPORTUNITIES OF INTERVENTION

LE CODE DE LA RUE ...



... LA RUE POUR TOUS

SI VOUS ÊTES CONDUCTEUR...

SI VOUS ÊTES PIÉTON...

SI VOUS ÊTES CYCLISTE...

SI VOUS UTILISEZ DES PATINS À ROULETTES OU UNE TROTINETTE...

SI VOUS ÊTES CYCLOMOTORISTE...

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

105

 universidade de aveiro

OPORTUNITIES OF INTERVENTION

Several alternatives for a sustainable mobility

- To promote distance communication (ex. tele-working, videoconference)
- To promote active modes
- To facilitate intermodality
- To improve public transportation
- To change the use and limit the abuse of private car
- To disseminate new technologies (ITS)
- To adapt taxation, pricing and use of standards to the desired behaviors
- To enforce established rules (parking, speed limits, ...)

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

106

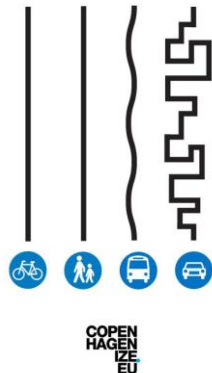
 universidade de aveiro

OPORTUNITIES OF INTERVENTION

(Re)guide public policies and involve stakeholders

- National public authorities
- Local authorities
- Transport operators
- businesses
- Socio- professional association
- NGOs
- Citizens
- Research units
- Vehicle and fuel industry
- Energy industry

Copenhagenize Traffic Planning Guide



COPENHAGENIZE EU

Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

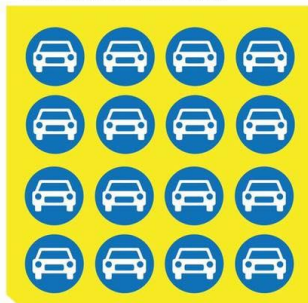
107

 universidade de aveiro

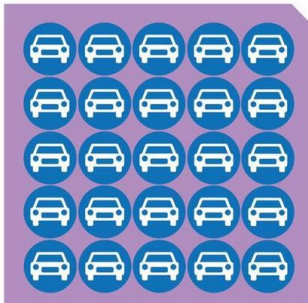
Life with cars.



Life with electric cars.



Life with autonomous cars.



Life in a modern city.



Competências Transferíveis II - Mobilidade Sustentável
Margarida Coelho – margarida.coelho@ua.pt

108